

Study guide

- **Read** Saracino, §6 and §8.
- Know the statement and proof of *Euclid's lemma* (Theorem 4.3). I also suggest knowing the upgrade that I called “Euclid+” in class.
- Understand the formula $o(x^m) = o(x)/(m, o(x))$ (Theorem 4.4(iii)) and its proof.
- An important special case: if $d \mid o(x)$, then $o(x^d) = o(x)/d$.
- Know the bijection between subgroups of a cyclic group G of order n and the divisors of n .
- Definition: the direct product $G \times H$.
- **Suggested (optional) problems** (answers to starred problems are in the back of the book):
 - §6: 1*, 2*
 - §8: 4, 8, 10(a)

1. This is a small enrichment exercise, with low stakes. You don't need to spend more than ten minutes on it, and a good faith effort will receive full points. Look up, wherever you like, a description of the “gauge group of the standard model of particle physics.” You should find a direct product $G_1 \times G_2 \times G_3$ of three groups. Look up and state a definition of each of the three groups. Optionally, explore a bit further and identify which aspects of physics (particles, forces, etc.) are explained by each of these three groups.
2. For each of the three order-8 groups $\mathbb{Z}_8, \mathbb{Z}_4 \times \mathbb{Z}_2$, and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$, determine the maximum order of an element in the group. Explain why this shows, as claimed in class, that the groups are not “the same up to relabeling” (or, as we'll say soon, they are not *isomorphic*).
3. Prove the following assertion, made in class: if G is a group and $g \in G$ has order $o(g) = 3$, then for any $h \notin \langle g \rangle$, the six elements $\{e, g, g^2, h, hg, hg^2\}$ are all distinct. Deduce that if G is a group of order 4 or 5, then it has no elements of order 3.

Note Soon, we'll prove *Lagrange's theorem*, which gives a much quicker proof of the claim above. But you should not use that theorem here; give a more elementary argument.

4. Let G, H be groups. Suppose that A is a subgroup of G , and B is a subgroup of H . Prove that $A \times B$ is a subgroup of $G \times H$.
5. Suppose that G, H are groups such that $G \times H$ is cyclic. Prove that G and H are both cyclic.
6. (a) Let G, H be groups. Show that $G \times H$ is abelian if and only if both G and H are abelian.
(b) Construct a nonabelian group of order 16, and one of order 24.
7. (Moved from Problem Set 3) Find all subgroups of Q_8 (the group of unit quaternions).

Note The group Q_8 is defined in Example 9 on page 47-8 of Saracino, which you should read before working on this problem.

8. (a) Let $f \in S_n$ be the cycle $(x_1, x_2, \dots, x_r)$. Show that $o(f) = r$.
(b) Suppose that $f = (x_1, x_2, \dots, x_r) \circ (y_1, y_2, \dots, y_s)$. Assume that these are *disjoint* cycles (that is, $x_i \neq y_j$ for all i, j). Prove that the order of f is the least common multiple of r and s .

(c) Find two transpositions whose product has order 3. This shows that the “disjoint” hypothesis is essential in part (b).

9. Suppose that g and h are elements of a group G , such that the following three conditions hold.

- $gh = hg$ (i.e. “ g and h commute”).
- $\langle g \rangle \cap \langle h \rangle = \{e\}$.
- g and h have finite order.

Prove that $o(gh)$ is the least common multiple of $o(g)$ and $o(h)$.

Hint Try a similar approach to problem 1 on Problem Set 3.